1. Introduction to Data Structure

Data Structure is a way of organizing, storing, and managing data efficiently so that it can be accessed and modified easily. It helps programmers write faster and more optimized programs.

2. Types of Data Structure

Data structures are mainly divided into two types: linear and non-linear. Linear data structures store data sequentially, such as arrays, stacks, queues, and linked lists. Non-linear data structures store data in a hierarchical or connected way, such as trees and graphs.

3. Array

An array is a linear data structure that stores multiple elements of the same type in continuous memory locations. Each element can be accessed using an index number.

4. Stack

A stack is a linear data structure that follows the LIFO principle, which means Last In, First Out. The element inserted last will be removed first. Common stack operations are push, pop, and peek.

5. Queue

A queue is a linear data structure that follows the FIFO principle, which means First In, First Out. The element inserted first will be removed first. Common queue operations are enqueue and dequeue.

6. Linked List

A linked list is a linear data structure where each element is called a node. Each node contains data and a pointer to the next node. It allows dynamic memory allocation.

7. Tree

A tree is a non-linear data structure that represents data in a hierarchical form. It consists of nodes connected by edges. The first node is called the root node.

8. Graph

A graph is a non-linear data structure made up of vertices and edges. It is used to represent networks such as social media connections, maps, and communication systems.

9. Importance of Data Structure

Data structures are important because they improve program efficiency, reduce memory usage, and make data handling easier. Choosing the correct data structure helps solve problems faster.

10. Applications of Data Structure

Data structures are used in databases, operating systems, search engines, networking, artificial intelligence, and web applications. They are very important in software development.